

University Interscholastic League
2023 – 2024 Junior High Number Sense Test C

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

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| <p>(1) $34 + 105 =$ _____</p> <p>(2) $13 \times 4 =$ _____</p> <p>(3) $95 - 42 =$ _____</p> <p>(4) $336 \div 3 =$ _____</p> <p>(5) $\frac{7}{8} - \frac{1}{4} =$ _____</p> <p>(6) $20 \div 0.5 =$ _____</p> <p>(7) $11 \times 44 =$ _____</p> <p>(8) $12^2 =$ _____</p> <p>(9) $8 + 8 \div 4 =$ _____</p> <p>*(10) $224 + 2024 + 2042 =$ _____</p> <p>(11) $41 \times 39 =$ _____</p> <p>(12) If the arithmetic mean of x, 10 and 9 is 13, $x =$ _____</p> <p>(13) Which is smaller: $\frac{6}{13}$ or $\frac{1}{2}$? _____</p> <p>(14) $8 \times 11 - 3 \times 11 =$ _____</p> <p>(15) $26 \times 86 =$ _____</p> <p>(16) $100 - 22 - 18 =$ _____</p> <p>(17) $55^2 =$ _____</p> <p>(18) $9 + 14 + 19 + 24 =$ _____</p> <p>(19) MMXXIV = _____ (Arabic Numeral)</p> | <p>*(20) $187.5 \times 480 =$ _____</p> <p>(21) $29 \times 31 + 1 =$ _____</p> <p>(22) $123 \times 101 =$ _____</p> <p>(23) If $f(x) = -2x^2 + 10$ then $f(-2) =$ _____</p> <p>(24) $6\frac{1}{10} \times 4\frac{1}{10} =$ _____ (mixed number)</p> <p>(25) $3\frac{7}{8} + 4\frac{3}{4} =$ _____</p> <p>(26) $13 - 13^2 =$ _____</p> <p>(27) The negative square root of 529 is _____</p> <p>(28) If n is to 8 as 3 is to 6, then $n =$ _____</p> <p>(29) If $12 + 5x$ is 42, then $x =$ _____</p> <p>*(30) $190^2 =$ _____</p> <p>(31) $42 \times 48 =$ _____</p> <p>(32) The sum of the prime numbers greater than 5 but less than 15 is _____</p> <p>(33) If $3.5 - 0.75 = n$, the $n^{-1} =$ _____</p> <p>(34) 12 is divisible by how many whole numbers? _____</p> <p>(35) 10 square feet = _____ square inches</p> <p>(36) 2 percent = _____ (common fraction)</p> <p>(37) If the sales tax is $8\frac{1}{2}\%$, what is the amount of sales tax on an item that costs \$100? \$ _____</p> |
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- (38) The LCM (18, 16) $\times$ GCF (18, 16) = _____
- (39) The area of an equilateral triangle with a side of 4-cm is $a\sqrt{3}$ and a = _____ cm²
- *(40) $111\sqrt{28900}$ = _____
- (41) 43 (base 6) = _____ (base 10)
- (42) $2^4 - 3^2$ = _____
- (43) 47×111 = _____
- (44) What is the area of a rhombus with diagonals 58 and 60? _____
- (45) $43 \text{ (base 6)} - 5 \text{ (base 6)}$ = _____ (base 6)
- (46) If $36 \div 9 = n^2 - 21$, and $n > 0$, then n = _____
- (47) If $A = \{0, 1, 2, 3\}$, $B = \{1, 2, 3, 4\}$ and $C = \{2, 4, 0\}$, how many elements are in $A \cap C \cap B$? _____
- (48) 32 pints = _____ quarts
- (49) The surface area of a rectangular solid that measures 4 by 2 by 3 is _____
- *(50) 320×624 = _____
- (51) In the sequence: 2, 4, a , 8, 10, b , 14, . . .
 $(b + a)^2$ = _____
- (52) What is the 6th term in the sequence: 2, 4, 6, . . .? _____
- (53) Each exterior exterior angle for a equilateral triangle is _____^o
- (54) What is the area of a parallelogram with opposite sides each 20 and altitude between them 15? _____
- (55) What is the area of a rectangle with a perimeter of 24 and length 9? _____
- (56) What is the radius of a circle with a circumference of 36π ? _____
- (57) The hypotenuse for a triangle with sides 12, 9 is _____
- (58) $(23^2 + 18 \times 4) \div 6$ has a remainder of _____
- (59) If the surface area for a cube is 54, its edge is _____
- *(60) $12^4 \times 101 \div 4^3$ = _____
- (61) $(4^{-1} \times 2^3)^{-1}$ = _____
- (62) 126 feet = _____ yards
- (63) 54 km/hr = _____ meters/sec
- (64) $25\% + 62\frac{1}{2}\%$ = _____ (common fraction)
- (65) $23^3 \div 6$ has a remainder of _____
- (66) $\frac{1+2+3+\dots+5}{2+4+6+\dots+10}$ = _____
- (67) The sum of the positive divisors of 15 is _____
- (68) $3!$ = _____
- (69) $2\frac{1}{4} \div \frac{3}{8}$ = _____
- *(70) 25 miles = _____ yards
- (71) $13^2 + 26^2$ = _____
- (72) $54\frac{6}{11}\%$ = _____ (common fraction)
- (73) $0.2333\dots$ = _____ (common fraction)
- (74) The volume of a sphere with a radius of 3 is $k\pi$ and k = _____
- (75) What are the odds of randomly picking a blue ball from a bag of 30 red and 24 blue balls? _____
- (76) $20^2 \times 25$ = _____
- (77) If the shortest distance between the points (12, 0) and (0, x) is 15, what is x ? _____
- (78) $11 \times \frac{5!}{3!}$ = _____
- (79) $(2.5)^2 - (3.5)^2$ = _____
- *(80) 24 hours = _____ seconds

2023 – 2024 University Interscholastic League Junior High Number Sense Test C – Key

(1) 139	*(20) 85500 – 94500	(38) 288	(59) 3
(2) 52	(21) 900	(39) 4	*(60) 31088 – 34360
(3) 53	(22) 12423	*(40) 17927 – 19813	(61) $\frac{1}{2}$; .5
(4) 112	(23) 2	(41) 27	(62) 42
(5) $\frac{5}{8}$; .625	(24) $25\frac{1}{100}$	(42) 7	(63) 15
(6) 40	(25) $8\frac{5}{8}$; 8.625; $\frac{69}{8}$	(43) 5217	(64) $\frac{7}{8}$
(7) 484	(26) -156	(44) 1740	(65) 5
(8) 144	(27) -23	(45) 34	(66) $\frac{1}{2}$; .5
(9) 10	(28) 4	(46) 5	(67) 24
*(10) 4076 – 4504	(29) 6	(47) 1	(68) 6
(11) 1599	*(30) 34295 – 37905	(48) 16	(69) 6
(12) 20	(31) 2016	*(50) 189696 – 209664	*(70) 41800 – 46200
(13) $\frac{6}{13}$	(32) 31	(51) 324	(71) 845
(14) 55	(33) $\frac{4}{11}$	(52) 12	(72) $\frac{6}{11}$
(15) 2236	(34) 6	(53) 300	(73) $\frac{7}{30}$
(16) 60	(35) 1440	(54) 300	(74) 36
(17) 3025	(36) $\frac{1}{50}$	(55) 27	(75) $\frac{4}{5}$; .8
(18) 66	(37) 8.50	(56) 18	(76) 10000
(19) 2024		(57) 15	(77) 9
		(58) 1	(78) 220
			(79) -6
			*(80) 82080 – 90720

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.